# SMART INDIA HACKATHON 2026

**Problem Statement ID -** 26231

**Problem Statement Title -**Digital Companion for Field Drug Testing

**Theme -** MedTech / BioTech / HealthTech

**PS Category -** Software

**Team ID -** K26234

**Team Name -** HexTech

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# PROPOSED SOLUTION

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## Problem

India: 1,20,010 NDPS cases in 2023 (source: NCRB, Crime in India 2023)

- Officers mainly judge the kit's colour change by looking at it, so lighting and personal judgement can affect the result.
- There is no standard colour value to compare the test result with.
- The test does not always leave a clear digital record of who performed it, where, and when, making it harder to verify later.

## Solution

- Check the Test - The mobile app captures the colour change and helps check the result.
- Record the Test - Saves the result along with the time, location, and Inspector ID.
- Manage the Records - The admin dashboard manages Inspectors, test records, evidence, and reports.

## User Experience

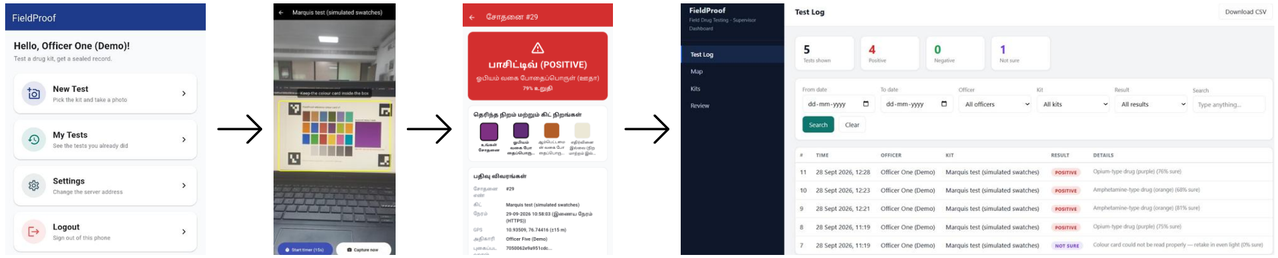
- **4 steps** - choose kit, take photo inside the box, see result, it gets saved. One test takes around 16 seconds (we checked in 19 scans).
- **Bad photo?** - app says NOT SURE, tells the reason (blurry, shiny or card not seen) and asks to take the photo again.
- **Easy to read** - Result is shown in big coloured letters, so anyone can read it easily. Works in Tamil too.

**Prototype Demo Video:** Prototype Video Link

## Our Approach

Test Kit → Take Photo → Check Colour → Get Result → Save Proof → Admin Review

**Prototype Flow -** real screenshots of our working prototype




*Result screen shown in Tamil. Test Log (right) can search and filter by date, officer, kit and result, and download CSV or PDF with QR.*

## Novelty

Eye check is guesswork, lab is slow, **FieldProof gives result on the spot for ₹10 with proof**

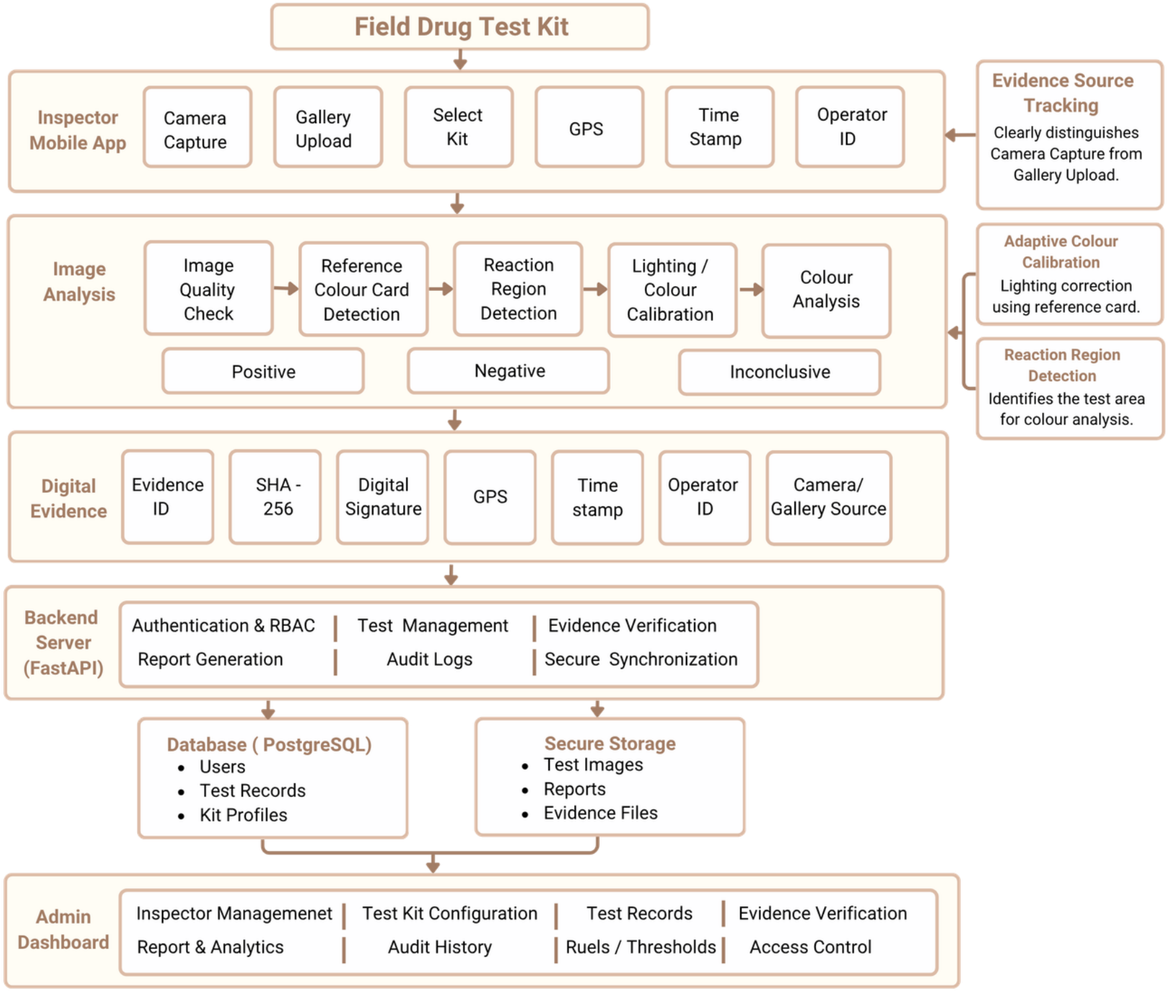
- **Handles Lighting Variations** - Uses a low-cost colour card to improve colour detection under different lighting.
- **Handles Unclear Results** - Blurry, shiny, or unclear images are marked INCONCLUSIVE instead of giving a wrong result.
- **Creates Verifiable Evidence** - Securely records the photo, GPS, time, and Inspector ID, with QR-based verification.
- **Supports Multiple Kits** - New kit colour profiles can be added from the Admin website without changing the app.

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# TECHNICAL APPROACH

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## System Architecture

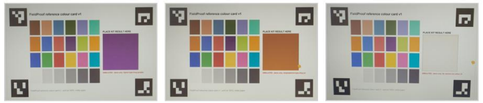



## Our Goals - Measured Results

- Testing: 37 computer made photos and 25 real phone scans.
- Accuracy: all 25 computer made photos correct. Real scans had 0 wrong answers (14 correct, others said NOT SURE).
- Colour Check: Used ΔE ≤ 14, 4-point colour matching, card-fit and glare thresholds.
- Speed: Result secured in 2.5 s typically, 5 s maximum.
- Tamper Test: All 4/4 modified records were detected successfully.

*Kit colours are simulated swatches (test photos + card on a screen), real kits need an FSL calibration trial.*
*Right now it is a lab prototype (TRL 4) tested on 1 phone. Next we test real kits with FSL on more phones.*

### Proof from Real Phone Scans (29 Sep 2026)



Purple card
**POSITIVE 79%**



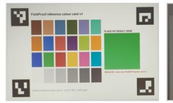
Orange card
**POSITIVE 76%**



Plain card
**NEGATIVE 75%**



In-between card
**NOT SURE**



Green card (not in kit)
**NOT SURE**



Blurry photo
**NOT SURE, retake**

We scanned using our app on 1 phone, in 1 room with window light. Card was shown on a laptop screen. We checked each card by eye, not using the app.

**Colour gap (Delta E), our limit is 14**
Right colour: 4.7 to 8.9
Other kit colours: 34 to 42
Trick cards: 16 to 31, so app said NOT SURE

**All 19 real scans**

| Card vs app | POSITIVE | NEGATIVE | NOT SURE |
|---|---|---|---|
| Positive card (8) | 5 | 0 | 3 |
| Plain card (4) | 0 | 4 | 0 |
| Trick card (7) | 0 | 0 | 7 |

**0 wrong answers**

All 9 clear scans correct
All 7 trick cards said NOT SURE
3 blurry photos, app asked to retake
Around 16 seconds per test (1 team member)

## Technology Stack

- Flutter (Dart)
- FastAPI (Python)
- OpenCV + NumPy
- CIELAB + ΔE2000
- PostgreSQL + SQLAlchemy
- React + Vite
- JWT + bcrypt
- SHA-256
- Ed25519
- NTP
- Leaflet + OpenStreetMap
- ReportLab

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# FEASIBILITY AND VIABILITY

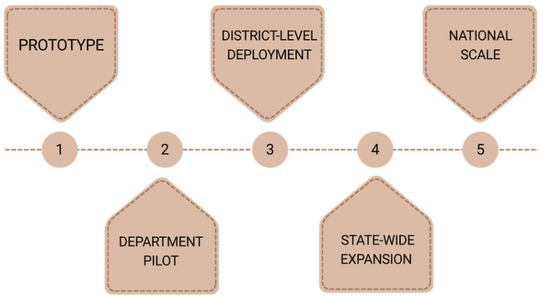
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## Feasibility & Sustainability

**Legal:** the app gives a first check only, the FSL lab confirms it.
**Privacy:** no face or suspect name is stored, only a sample code.

- **Technical** - Core app tested with OpenCV-based colour analysis.
- **User-Friendly** - Capture one photo and get a result within seconds.
- **Low Cost** - Uses an existing Android phone, a reusable ~₹10 colour card, and open-source tools.
- **Operational** - Simple photo-based workflow with minimal training required.
- **Scalable** - Add new test-kit colour profiles through the Admin website without updating the app.
- **Sustainable** - Government-cloud hosting, regular profile updates, and secure data storage support long-term use.

## Phased Implementation & Scalability




### Pilot Plan (proposed)

- **Where** - 1 district in Tamil Nadu, with local police and the nearest FSL
- **Who** - 20 officers, for 3 months
- **Check** - every field result is checked again in the FSL lab
- **Success** - 95% or more same as lab, NOT SURE below 10%, every record passes the check
- **Cost** - 20 cards at ₹10 = ₹200, server ₹3,000 a month for 3 months = ₹9,000. Total about ₹9,200 (estimate)

## Potential Challenges & Strategies

| Real-world problem | What our app does |
|---|---|
| Bad light (street lamp / torch) | Colour card fixes the colours |
| Shiny or blurry photo, card missing | Says NOT SURE and asks to retake |
| Colour is in between two results | Says NOT SURE; senior officer checks |
| Fake photo or edited record | In-app camera only + photo fingerprint |
| Different kits and phones | Kit settings + colours fixed every photo |

## Cost & Deployment Plan

**After the pilot,** the cost can come from the state police or NCB IT budget.
No licence fee as it is open source, so running cost stays low.

| COST | HOSTING | MAINTENANCE | DEPLOYMENT | SCALE |
|---|---|---|---|---|
| Uses an existing phone, ₹10 reusable colour card, and open-source software. | Runs on government cloud for the app, database, and admin dashboard. | Regular security updates and easy kit-profile updates as needed. | Start with a pilot, test in the field, then expand deployment. | New test kits can be added through the website without updating the app. |

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# IMPACT AND BENEFITS

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## Inspector Benefits

- Faster, more consistent testing
- Less manual colour checking
- INCONCLUSIVE when unclear
- Easy digital test records
- Quick access to past test records

## Organisation Benefits

- Proof of every test: time, place, officer & photo
- Secure and verifiable records
- Less paperwork with PDF & QR
- Easy search, reports and monitoring
- Better field-test documentation

## QR-Based Record Verification

**FieldProof - Field Test Record**

Presumptive field result — not a substitute for laboratory confirmatory testing.

**POSITIVE**
Opium-type drug (purple) | 76% sure



Scan to check this record
http://127.0.0.1:8000/v/11

| | |
|---|---|
| Test number | 11 |
| Test ID | 4878cf1d-8f00-4f66-9f73-30c50ed39e62 |
| Saved at (UTC) | 2026-09-28T06:58:23.601Z (time from: internet time (HTTPS)) |
| Phone time | 2026-09-28T12:28:21+05:30 (difference: -2.6 s) |
| GPS location | 10.935299, 76.743282 (±14.791999816894531 m) |
| Officer | Officer One (Demo) (user id 1) |
| Device ID | android-1e42a9c170701656ac8a22b0ca44a6e0 |
| Kit | Marquis test (simulated swatches) [DEMO-MARQUIS-SIM], profile version 1 |
| Lab sample ID | - |
| Review status | NOT_NEEDED |

**FieldProof · Record check**

**✓ Record is genuine**
The photo and details were not changed after they were saved.

**What we checked**

- ✓ The photo is the same as when it was taken
- ✓ The saved details match their fingerprint (hash)
- ✓ The FieldProof server's digital signature is valid
- ✓ The record is correctly linked to the record before it
- ✓ The result, officer and kit shown are the ones that were signed

**Record details**

| | |
|---|---|
| Test number | 11 |
| Result | POSITIVE |
| What the kit showed | Opium-type drug (purple) |
| How sure | 76% |
| Kit | Marquis test (simulated swatches) (version 1) |
| Officer | Officer One (Demo) |
| Saved at (UTC) | 2026-09-28T06:58:23.601Z (internet time (HTTPS)) |
| GPS | 10.93530, 76.74328 |
| Test ID | 4878cf1d-8f00-4f66-9f73-30c50ed39e62 |
| Photo fingerprint (SHA-256) | 294b35454c7e963c7e0f939b4f0e9cd10a279e69fca62634d61ea030d98a9e57 |

## Map - Geo-Tagged test result




**Court Certificate:** every PDF report now has a filled BSA section 63 page (photo hash, phone, time, GPS) for the officer and expert to sign. Open a full sample report (PDF, 2 pages)

## Tamper Test - we tried it on our saved records

**Test #10**
prev 2ece5c80
hash 6b1b4e7f

→ **Test #11**
prev 6b1b4e7f
hash 03dd9f0c

→ **Test #12**
prev 03dd9f0c
hash a110cff6

Ed25519 signature of test 11
JRIdu6ERgadC+2n6chKfUStp...
Photo SHA-256
294b35454c7e963c7e0f939b...

| What we changed (test #11) | Check page | Failed check |
|---|---|---|
| Nothing (original) | GENUINE | none, all 5 passed |
| 1 byte of the photo | TAMPERED | photo |
| Result inside sealed copy | TAMPERED | hash, signature, fields |
| Result in database row | TAMPERED | fields |
| The signature | TAMPERED | signature |

We tried this on our real saved records on 29 Sep 2026, only in memory. Each test keeps the hash of the test before it.

## Future Scope (our plan after SIH)

- Offline Mode (month 1) - save the test on phone, send it when network is back.
- Phone Security (month 2) - sign each test with the phone's security chip.
- Police System Integration (after pilot) - apply to NIC or MHA to link with e-Sakshya and CCTNS. Our CSV export is ready.
- Lab Comparison (during pilot) - compare field results with FSL lab results.
- Regional Languages - Tamil is done in the app. Hindi and voice help in month 3.

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# RESEARCH AND REFERENCES

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| What they found | How we use it | Research |
|---|---|---|
| **IEEE EMBC 2016**<br>Phone camera colours change with light and phone model; reference colours can help handle this. | Our ₹10 colour card fixes colours in every test | [1] H. Karisen and T. Dong, "Illumination and device independence for colorimetric detection of urinary biomarkers with smartphone," in Proc. 38th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC), Orlando, FL, USA, 2016, pp. 5184-5187, doi: 10.1109/EMBC.2016.7591895. |
| **IEEE TPS-ISA 2024**<br>Digital evidence must maintain integrity and show how it was handled. | Photo fingerprint + signature + chained log | [2] S. Nath, K. Summers, J. Baek, and G.-J. Ahn, "Digital evidence chain of custody: Navigating new realities of digital forensics," in Proc. IEEE 6th Int. Conf. Trust, Privacy Secur. Intell. Syst. Appl. (TPS-ISA), Washington, DC, USA, 2024, pp. 11-20, doi: 10.1109/TPS-ISA62245.2024.00012. |
| **Forensic Science International 2014**<br>A mobile phone camera can measure drug-test colours using RGB values. | Our app reads the test colour as a number | [3] A. Choodum, K. Parabun, N. Klawach, N. Nic Daeid, P. Kanatharana, and W. Wongniramaikul, "Real time quantitative colourimetric test for methamphetamine detection using digital and mobile phone technology," Forensic Sci. Int., vol. 235, pp. 8-13, Feb. 2014, doi: 10.1016/j.forsciint.2013.11.018. |
| **Analytical Chemistry 2016**<br>By-eye colour interpretation is subjective; smartphone image analysis can provide objective analysis. | App gives POSITIVE / NEGATIVE / NOT SURE | [4] S. T. Krauss et al., "Objective method for presumptive field-testing of illicit drug possession using centrifugal microdevices and smartphone analysis," Anal. Chem., vol. 88, no. 17, pp. 8689-8697, Sep. 2016, doi: 10.1021/acs.analchem.6b01982. |
| **Digital Discovery 2023**<br>Computer vision can quantify and compare colour changes that may look similar by eye. | App shows photo colour next to kit colours | [5] N. Bugeja et al., "Teaching old presumptive tests new digital tricks with computer vision for forensic applications," Digital Discovery, vol. 2, no. 4, pp. 1143-1151, 2023, doi: 10.1039/D3DD00066D. |